VoltTwin AI

SMARTER DATA • HEALTHIER BATTERIES • BETTER TOMORROW

PREDICT BATTERY HEALTH WITH AI



AI-Powered Battery Health and Remaining Useful Life Prediction

VoltTwin AI leverages Machine Learning to analyze battery cycle, voltage, temperature, capacity and State of Health to predict Remaining Useful Life with high accuracy.

Start Prediction

View Dashboard

Battery Health
Monitor the current health of your battery in real-time.

RUL Prediction
Predict Remaining Useful Life with advanced ML models.

Machine Learning
AI models learn from data to deliver accurate predictions.

Battery Health Prediction
Get accurate insights into the State of Health of your battery.

RUL Prediction
Predict the Remaining Useful Life and plan maintenance better.

AI-Powered Analysis
Advanced Machine Learning algorithms for smarter decisions.

Battery Monitoring
Continuous monitoring for safer, longer and smarter usage.

HOW IT WORKS

Battery Data
Collect cycle, voltage, temperature, capacity and SOH data.

Machine Learning
Train AI models using historical battery performance.

Health Analysis
Analyze battery health and performance patterns.

RUL Prediction
Predict Remaining Useful Life with high accuracy.

AI TODAY. RELIABLE ENERGY TOMORROW.

VoltTwin AI

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Predict Battery Health with AI

Machine Learning based Battery Health and Remaining Useful Life Prediction

VoltTwin AI analyzes battery cycle, voltage, temperature, capacity and State of Health to predict Remaining Useful Life.

Start Prediction | View Dashboard



- Battery Health Prediction
- RUL Prediction
- AI-Powered Analysis
- Battery Monitoring

Battery Data → Machine Learning → Health Analysis → RUL Prediction


VoltTwin AI
Dashboard
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Help & Support
Help & Support
Dashboard
AI Model Status: Online
Last Updated: Just now
INPUT PARAMETERS
Cycle Count: 500
Voltage: 3.85 V
Temperature: 30 °C
Capacity: 1.75 Ah
State of Health: 88%
PREDICT BATTERY HEALTH
PREDICTION RESULTS
Predicted Remaining Useful Life (RUL): 9.32 Cycles
Battery Status: GOOD
Battery Health: 92%
Confidence Score: 98.7%
Degradation Rate: 0.87% /cycle
Power Efficiency: 87.4%
Temperature Status: Optimal
BATTERY HEALTH OVERVIEW
92%
Health Score
Health Level: Excellent
Risk Level: Low Risk
Estimated Life Left: 9.32 Cycles
RUL Prediction Trend
100
75
50
25
0
Time
Time
Time
Time
Time
Thme
Feature Importance
40
30
20
10
0
State of Health (SOH)
Temperature – 15%
Voltage – 11%
Capacity – 8%
Cycle Count – 42%
State of Health (SOH) – 24%
Temperature – 15%
Voltage – 11%
Capacity – 8%


5:28 PM
VoltTwin AI
Battery Health Prediction
Home
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Settings
Enter Battery Parameters
Cycle Count
500
Voltage (V)
3.85
Temperature (°C)
30
Capacity (Ah)
1.75
State of Health (%)
88
PREDICT BATTERY HEALTH
Prediction Result Summary
Predicted Remaining Useful Life
9.32 Cycles
Battery Status
GOOD
92%
Health Score
92%
Confidence
98.7%
Based on the provided input, the battery shows excellent health with high predictive confidence.


VoltTwin AI
Battery Health Prediction Result
PREDICTION RESULT
MAIN RESULT CARD
SECONDARY RESULT CARD
Confidence Score
98.7%
Power Efficiency
87.4%
Predicted Remaining Useful Life (RUL)
9.32 Cycles
Battery Status: GOOD
Battery Health: 92%
Health Score
92%
Excellent
Low Risk
Degradation Rate
0.87% /cycle
Temperature Status
Optimal
SECONDARY RESULT CARD
INPUT SUMMARY
INPUT SUMMARY
Cycle Count: 500
Voltage: 3.85 V
Temperature: 30 °C
Capacity: 1.75 Ah
SOH: 88%
Prediction Summary
The battery is currently in good condition with an estimated Remaining Useful Life of 9.32 cycles.
Predict Again
View History

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VoltTwin AI

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About VoltTwin AI

Battery Health Prediction Using Machine Learning

Project Objective

Predict the Remaining Useful Life (RUL) of lithium-ion batteries using machine learning.

Process Overview

Input Parameters
Cycle, Voltage, Temperature, Capacity, SOH

Output
Predicted Remaining Useful Life (RUL)

Technology Stack

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Random Forest Regressor

AI Model

Model:

Random Forest Regressor

Model Performance

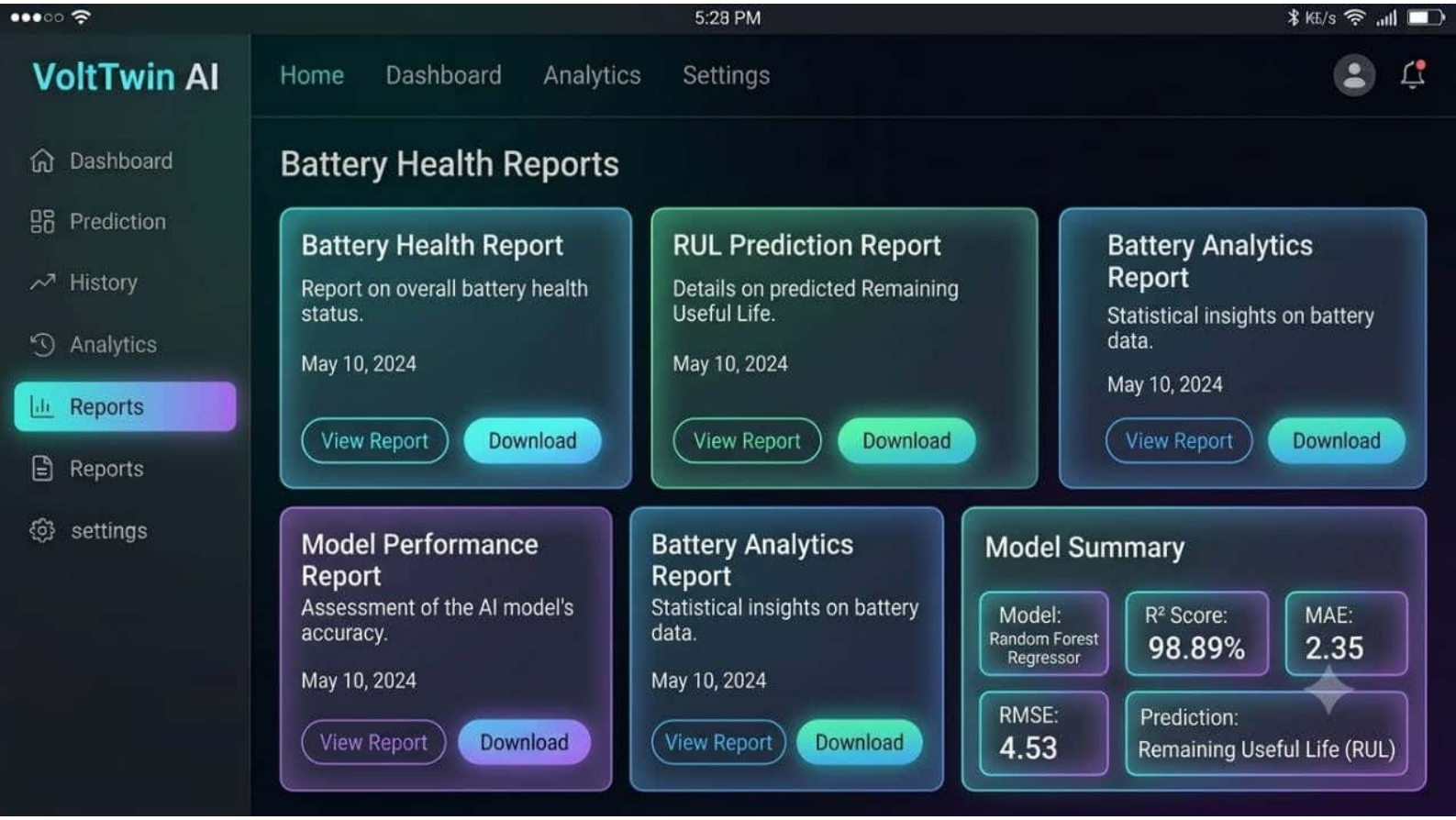
R^2 Score:	MAE:
98.89%	2.35

RMSE: 4.53



AI

Data Input → Feature Selection → Model Training (RF) → RUL Prediction → Health Report


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Battery Health Reports
Battery Health Report
Report on overall battery health status.
May 10, 2024
View Report
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RUL Prediction Report
Details on predicted Remaining Useful Life.
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View Report
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Battery Analytics Report
Statistical insights on battery data.
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Download
Model Performance Report
Assessment of the AI model's accuracy.
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View Report
Download
Battery Analytics Report
Statistical insights on battery data.
May 10, 2024
View Report
Download
Model Summary
Model:
Random Forest Regressor
R² Score:
98.89%
MAE:
2.35
RMSE:
4.53
Prediction:
Remaining Useful Life (RUL)


5:28 PM
VoltTwin AI
Battery Analytics
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Voltage Distribution
Count
Cell Voltage [V]
SOH Distribution
Count
State of Health [%]
Voltage vs Temperature
Temperature [°C]
Voltage [V]
Capacity vs SOH
State of Health [%]
Capacity [Ah]
Voltage vs RUL
Remaining Useful Life [Cycles]
Voltage [V]
Capacity vs Cycle
Cycle Count
Capacity [Ah]
Correlation Heatmap
Cycle SOH Temp Volt Capa RUL
Cycle 1 0.76 0.33 0.95 0.78 0.23
SOH 0.73 1 0.95 0.73 0.97 0.73
Temp 0.23 0.75 1 0.88 0.34 0.53
Volt 0.95 0.73 0.83 1 0.79 0.33
Capa 0.78 0.33 0.43 0.25 0.56 1
Feature Importance
Random Forest Feature Importance
25.52
23.27
15.9
1.25
6.99
Cycle Count
State of Health (SOH)
Temperature
Voltage
Capacity
Total Records
1415
Features
5
Target
RUL
Model
Random Forest Regressor
R² Score
98.89%


5:28 PM
VoltTwin AI
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Prediction History
Search Predictions
Filter History
Export Report
Date & Time | Battery ID | Cycle | Voltage (V) | Temperature (°C) | Capacity (Ah) | SOH (%) | Predicted RUL | Status
(Real Date & Time) | B0005 | 120 | 3.45 | 31.2 | 1.62 | 91% | 48.6 | Good
(Real Date & Time) | B0005 | 200 | 3.41 | 32.1 | 1.55 | 87% | 31.4 | Good
(Real Date & Time) | B0005 | 300 | 3.35 | 33.0 | 1.48 | 82% | 20.7 | Moderate
(Real Date & Time) | B0005 | 500 | 3.85 | 30.0 | 1.75 | 88% | 9.32 | Good
(Real Date & Time) | B0005 | 120 | 3.45 | 31.2 | 1.62 | 91% | 48.6 | Good
(Real Date & Time) | B0005 | 200 | 3.41 | 32.1 | 1.55 | 87% | 31.4 | Moderate
(Real Date & Time) | B0005 | 300 | 3.35 | 33.0 | 1.48 | 82% | 20.7 | Good
(Real Date & Time) | B0005 | 500 | 3.85 | 30.0 | 1.75 | 88% | 9.32 | Good

File Edit Selection View Go Run Terminal Help — VoltTwin AI – Battery Health Prediction

EXPLORER

- VOLTTWIN_AI
  - notebooks
    - VoltTwin AI – Battery Hea...
  - data
    - raw
      - rt.py
      - level_dataset.csv
      - network.py
      - notetook.csv
    - README.md
  - .gitignore
  - package.json
  - package-lock.json
  - README.md

OUTLINE

TIMELINE

Jupyter. Notebook

VoltTwin AI > Battery Health Prediction > notebook

+ Code | Run cell | Output in outsat | Python Notebook

VoltTwin AI – Battery Health Prediction

Dataset

```
df = pd.read_csv("../data/raw/Battery_cycle_level_dataset.csv")
df.head()
```

[1] ✓ Cell — Python

	le	voltage	temperature	capacity	soh	rul
1		3.532781	32.536891	1.861976	1.000000	167
2		3.542968	32.643595	1.851862	0.994568	166
3		3.553056	32.522526	1.840808	0.988631	165
4		3.545849	32.492083	1.850058	0.993599	164

Data Preprocessing

```
df.isnull().sum()
```

[2] — Python

```
battery_id      0
cycle           0
```

```
df.shape
```

[3] — Python

```
(1415, 7)
```

```
df.dtypes
```

[4] — Python

```
battery_id       str
cycle          int64
voltage      float64
temperature  float64
capacity     float64
soh          float64
rul            int64
```



Python (2.315 kernel) 0 ⚠ 0 — Ln 17, Col 12 — Spaces: 4 — UTF-8 — LF — 10 kernel